# Trends in Open Source AI Models (2025-2026)

## Project Overview:

This project presents a comprehensive analytical study of open-source AI model evolution using data sourced from the Hugging Face community over the past year. The objective of the study is to examine model popularity, download dynamics, geographic and organizational influence, and performance patterns across the open-source AI ecosystem.

Using descriptive statistics, distribution analysis, and multivariate visualizations, this study investigates adoption concentration, skewness in popularity distribution, growth volatility, and ecosystem-level structural drivers. By integrating both temporal and structural perspectives, the project aims to identify the factors associated with sustained model success and provide data-driven insights into broader trends shaping the open-source AI landscape.

## Data Source:

- **Dataset name :** Open Model Evolution Dataset
- **Source :** HuggingFace Community
- **File Type :** CSV
- **Number of Record : 19,361**
- **Number of Attributes : 19**
- **Year : 2025-2026**

- **Dataset Description:**The dataset contains unique model identifiers, model names, authors, creation timestamps, and lifecycle tracking variables, enabling temporal and structural analysis of open-source AI model development. It captures aggregated engagement indicators ,allowing evaluation of model popularity, adoption concentration, and ecosystem-level success patterns.The dataset includes organization type (user/team/company), country information (original and derived), model lineage, supported languages, and modality transformations (e.g., Text → Text, Image → Video), supporting analysis of geographic influence, derivation networks, and multimodal evolution trends.

## Problem Statement:

The rapid growth of open-source AI models has created a highly competitive and uneven ecosystem, where only a small fraction of models achieve significant visibility and adoption. However, there is limited structured analysis explaining what drives model popularity, growth acceleration, and sustained success across

countries and organizations. This project aims to investigate the distribution of downloads, lifecycle patterns, organizational influence, and geographic contributions within the Hugging Face open-source community. By analyzing both temporal growth dynamics and cross-sectional structural characteristics, the study seeks to identify concentration patterns and structural drivers of model performance. Ultimately, the problem addressed is understanding how model attributes and ecosystem factors influence adoption trends in the evolving open-source AI landscape.

## Objectives:

- Analyze AI model adoption trends using download metrics to identify the most popular and fast-growing models.
- Examine the geographic distribution of AI model development across countries and organizations.
- Compare model modalities (Text, Image, Audio, Video, Multimodal) and study their usage patterns.
- Identify the role of contributor types (users, teams, companies, free/community) in AI model creation.
- Explore language support trends, highlighting multilingual versus single-language models.
- Investigate model lineage and derivation, understanding how base models are reused or fine-tuned.
- Perform time-based analysis to observe model release patterns and ecosystem growth over time.

## Attribute Details

| Attribute Name | Data Type | Description |
|---|---|---|
| **id** | String (Object) | Unique identifier assigned to each AI model record. |
| **model** | String | Full model repository name including organization/user prefix. |
| **author** | String | Original creator or publishing entity of the model. |
| **created** | Datetime | |

| | | |
|---|---|---|
| | | Timestamp indicating when the model was initially released. |
| **time** | DateTime | Observation or snapshot date used for time-series tracking. |
| **downloadsAllTime** | Integer | Total cumulative number of downloads, representing model popularity. |
| **annotator** | Integer | Number of annotators or contributors associated with the model. |
| **access_tier** | Categorical | Access classification (e.g., user, team, free) defining model availability level. |
| **org_type** | Categorical | Type of organization publishing the model (user, team, company). |
| **org_country** | Categorical/List | Country associated with the publishing organization. |
| **derived_model_country** | Categorical | Country linked to the base or derived model origin. |
| **org_country_grouped** | Categorical | Country linked to the base or derived model origin. |
| **derived_country_grouped** | Categorical | Regionally grouped classification of the derived model's country |
| **derived_author** | String | Author or organization from which the model is derived. |
| **model_derived_from** | String/List | Parent or base model used for derivation or fine-tuning. |

| | | |
|---|---|---|
| **model_languages** | List(Categorical) | Languages supported or trained on by the model. |
| **model_modality** | Categorical | Input-output modality transformation (e.g., Text → Text, Image → Video). |

## Tools & Technologies:

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Plotly**
- **Google Colab**

## Data Pre-Processing and Feature Engineering

The raw dataset was initially stored as **df_raw**, preserving the original structure for reproducibility and auditability. The data was then systematically cleaned and transformed to ensure consistency, reliability, and analytical usability. Following the cleaning phase, the dataset was strategically split into two analytical layers: **df_master** for time-series and growth-based analysis, and **df_models** for cross-sectional evaluation of static model popularity and structural attributes. This separation enabled clearer modeling of lifecycle dynamics and aggregated performance behavior.

## Data Cleaning

- Renamed column names for improved clarity and interpretability.
- Standardized column data types (e.g., datetime conversion, integer casting, categorical formatting).
- Cleaned list-type categorical columns to ensure consistent list structures.
- Replaced missing values in the org_type column using mode imputation.

- Used the **Abstract Syntax Tree (AST)** library to clean the model_derived_from column and replaced ambiguous or null values with "Unknown" for consistency.

## Feature Engineering

### df_master (Time-Series Dataset)

- Computed lifecycle-based variables such as model_age_days, model_age_weeks, and model_week.
- Engineered growth-related metrics including weekly_growth.
- Imputed derived temporal columns to ensure continuity in panel data analysis.
- Structured the dataset to support longitudinal growth tracking and trend analysis.

### df_models (Cross-Sectional Dataset)

- Extracted unique model-level records for static popularity evaluation.
- Created structural indicator features such as has_known_lineage, same_author_derived, is_multimodal_input, and is_active_model.
- Aggregated performance metrics to enable comparative analysis across models, organizations, and countries.
- Structured the dataset for distributional analysis and ecosystem-level insights.

## Analysis and Visualization

### Exploratory Data Analysis:

Exploratory Data Analysis was conducted to understand the structural composition, distributional characteristics, and growth behavior of the dataset. The cross-sectional dataset, **df_models**, contains **550 rows and 17 columns**, representing unique AI models and their aggregated popularity and structural attributes. The time-series dataset, **df_master**, consists of **19,361 rows and 19 columns**, capturing longitudinal observations and growth dynamics across model lifecycles.

### Statistical Analysis:

Statistical analysis was conducted using descriptive statistics, distribution diagnostics, correlation analysis, and growth-based time-series evaluation. Measures such as mean, median, variance, skewness, and kurtosis revealed a highly right-skewed and heavy-tailed download distribution, indicating strong concentration effects. Time-series metrics, including weekly growth and acceleration, further highlighted lifecycle-dependent momentum patterns and periods of surge-driven adoption.

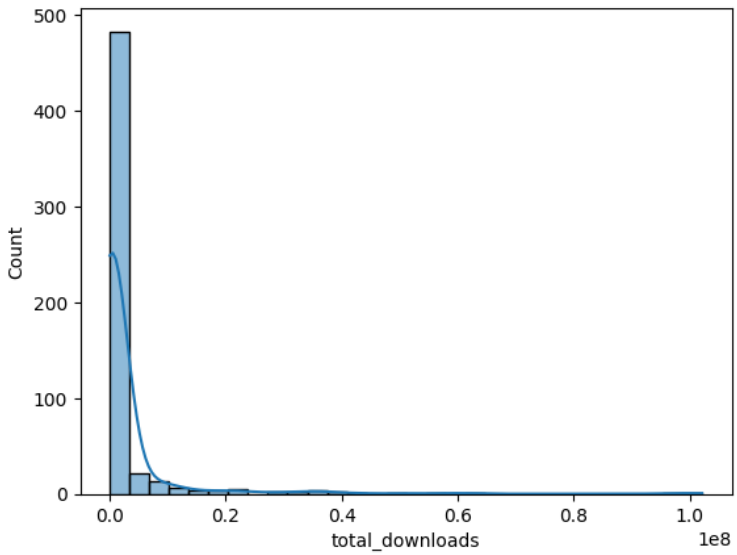
**Visualizations:**

Data visualization techniques were used to uncover patterns, trends, and structural relationships within the dataset. Histograms, boxplots, bar charts, scatter plots, heatmaps, and time-series line plots were employed to examine download distribution, organizational impact, country-wise performance, modality adoption, and lifecycle growth dynamics.

## Model Analysis

### 1.Univariant Analysis:

### Distribution of Total Downloads

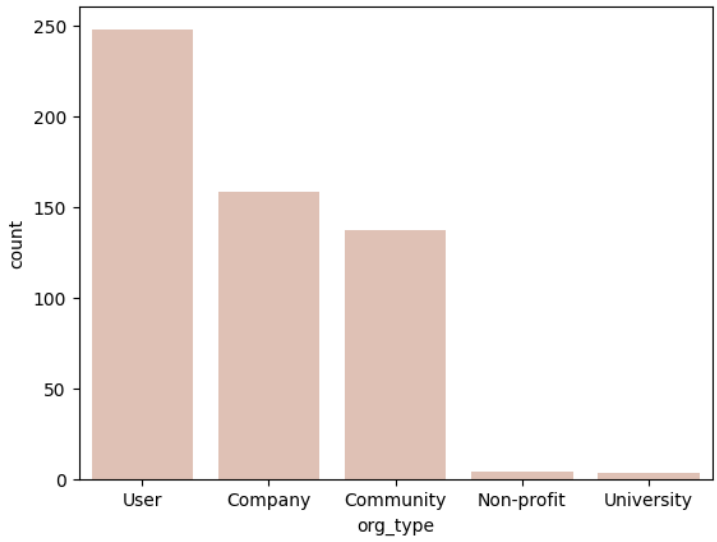



### Interpretation:

- The histogram shows an extremely right-skewed distribution, where the majority of models have relatively low download counts clustered near zero, while a small number of models extend into very high download ranges.

- The long right tail confirms the presence of significant outliers, reinforcing a winner-takes-all pattern in which only a few models capture disproportionately large user attention.

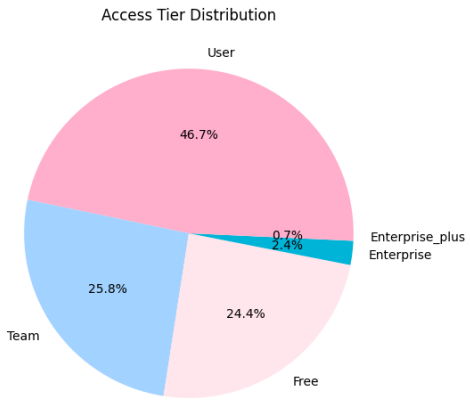
## Organization Type Distribution




## Interpretation:

- The chart indicates that most models are developed by individual users, followed by companies and community groups, showing that independent creators play a dominant role in model contributions.
- In contrast, non-profits and universities contribute very few models, suggesting limited institutional participation compared to individual and corporate contributors.

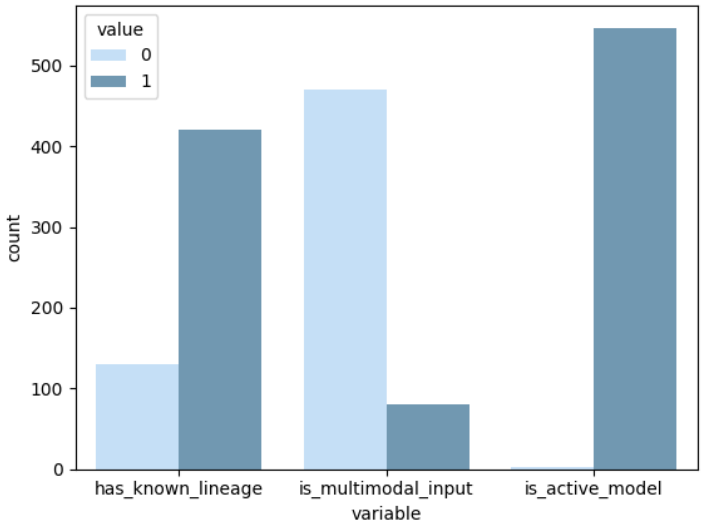
## Model Users Distribution



**Interpretation:**

- The distribution shows that individual consumers dominate usage (46.7%), indicating that nearly half of the models are primarily accessed at the individual level rather than organizational level.
- Enterprise adoption remains comparatively low (3.1% combined for Enterprise and Enterprise Plus), suggesting that large-scale corporate usage is still limited compared to consumer and team-based access.

**Boolean Features Distribution**

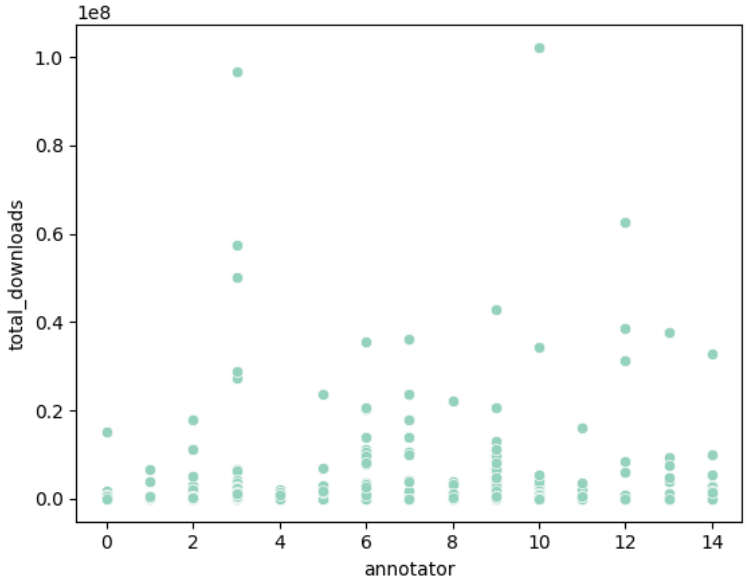



**Interpretation:**

- The chart shows that most models have a known lineage (True), indicating that derivation or model inheritance is common, whereas multimodal input support is relatively rare, with the majority of models being unimodal.
- Additionally, all models are active (100% True), suggesting the dataset is heavily skewed toward currently maintained or operational models rather than deprecated ones.

## 2. Bi-Variant Analysis

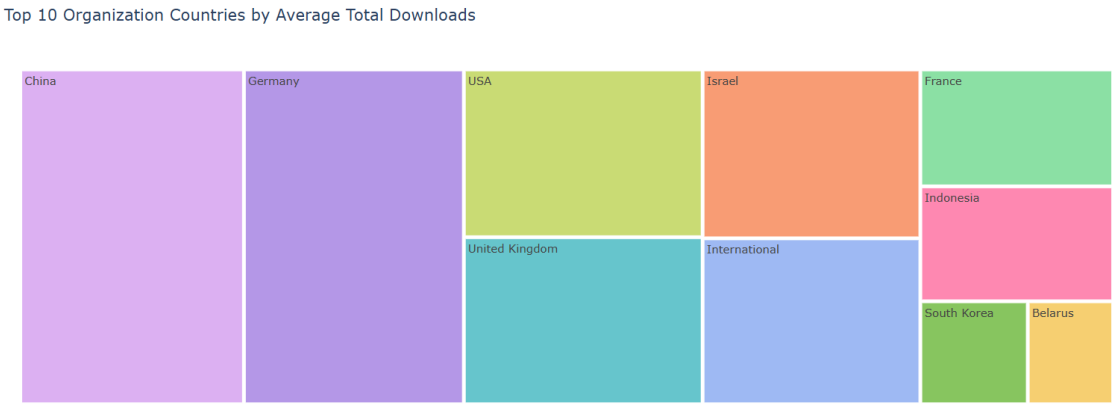
**Downloads vs Annotators**




**Interpretation:**

- The scatter plot suggests that models developed by teams of around 5 to 10 annotators tend to achieve higher download counts, indicating that moderately sized teams may contribute to stronger model performance.
- However, the presence of extreme outliers shows that while team size can be associated with higher downloads, success is not solely dependent on the number of annotators.

**Downloads vs Top Countries**

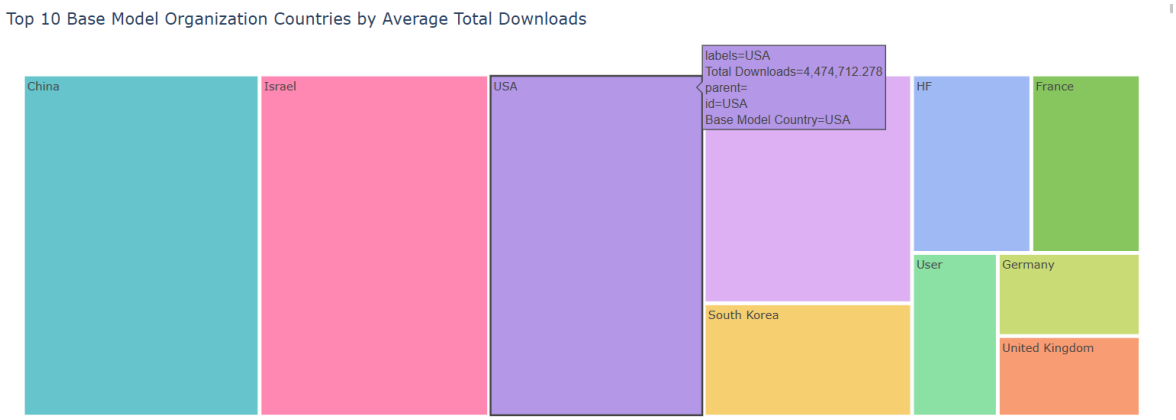



**Interpretation:**

- The chart indicates that China has the highest average downloads per model, followed by Germany, the United States, and the United Kingdom, suggesting strong model performance from these regions.

- This implies that certain countries may have stronger AI ecosystems or higher-performing flagship models that significantly boost their average download metrics.

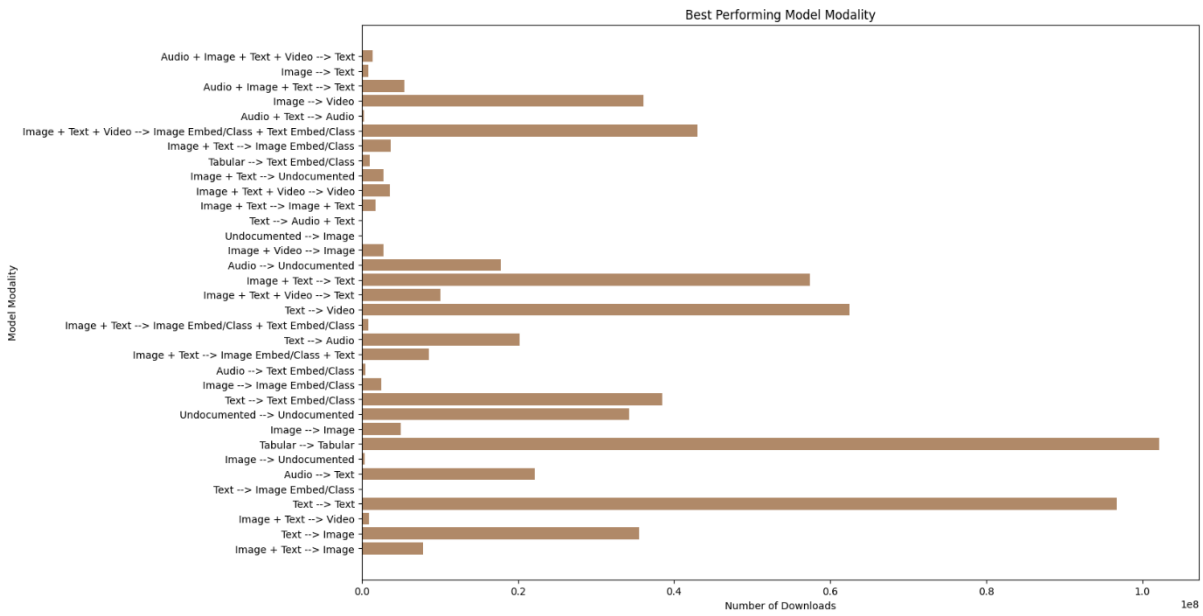
## Downloads vs Top Base Model Organization




## Interpretation:

- The chart shows that base models originating from China have the highest average downstream downloads, followed by Israel and the United States, indicating that foundational models from these regions tend to generate strong derivative performance.
- This suggests that the country of the original base model may influence the success of derived models, reflecting the global impact and quality of certain foundational AI ecosystems.

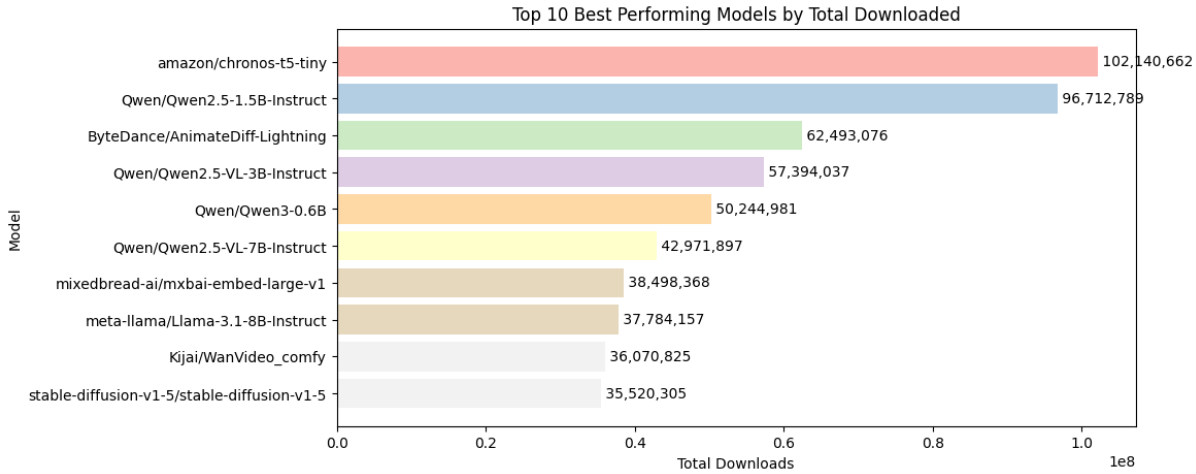
## Downloads vs Modal Modality



**Interpretation :**

- The chart indicates that Tabular-to-Tabular, Text-to-Text, Text-to-Video, and Image+Text-to-Text modalities receive the highest download volumes, suggesting strong demand for data transformation, generative text, and multimodal applications.
- This highlights that practical, widely applicable modalities—particularly those involving text processing and cross-modal generation—tend to dominate user adoption compared to more niche model types.

**Top 10 Best Performing Models**

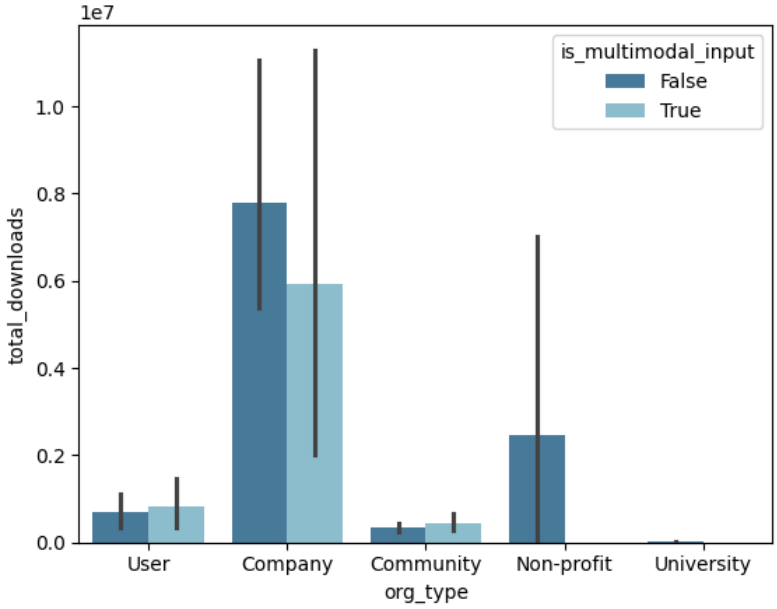



**Interpretation:**

- The bar chart shows that company-developed models without multimodal input support (single-input models) achieve the highest download volumes, indicating that simpler, specialized architectures may currently drive stronger adoption.
- Similarly, non-profits and users also show higher downloads for unimodal models, suggesting that single-modality models remain more widely used and commercially impactful than multimodal ones in this dataset.

## 3. Multivariant Analysis

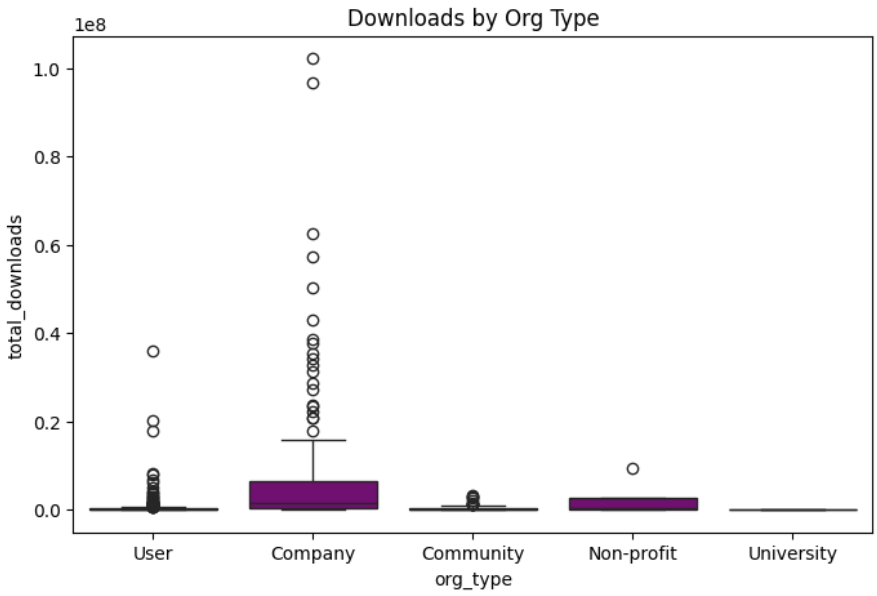
## Does multimodal input impact downloads?




**Interpretation:**

- The bar chart shows that company-developed models without multimodal input support (single-input models) achieve the highest download volumes, indicating that simpler, specialized architectures may currently drive stronger adoption.
- Similarly, non-profits and users also show higher downloads for unimodal models, suggesting that single-modality models remain more widely used and commercially impactful than multimodal ones in this dataset.

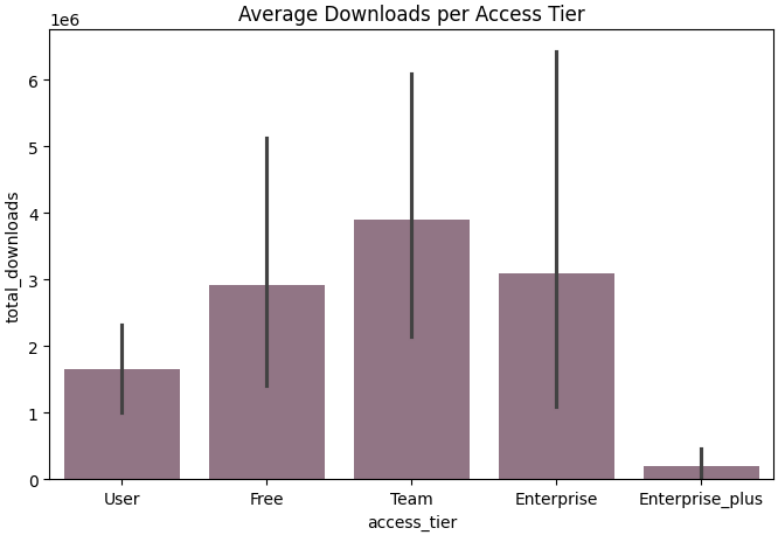
## Downloads vs Organization Type



**Interpretation**

- The boxplot indicates that company-developed models have the highest median downloads and the widest spread, along with numerous extreme outliers, suggesting that companies are more likely to produce blockbuster models.
- Individual users also show notable outliers but with comparatively lower central tendency, while non-profits demonstrate more limited dispersion and overall lower download performance.

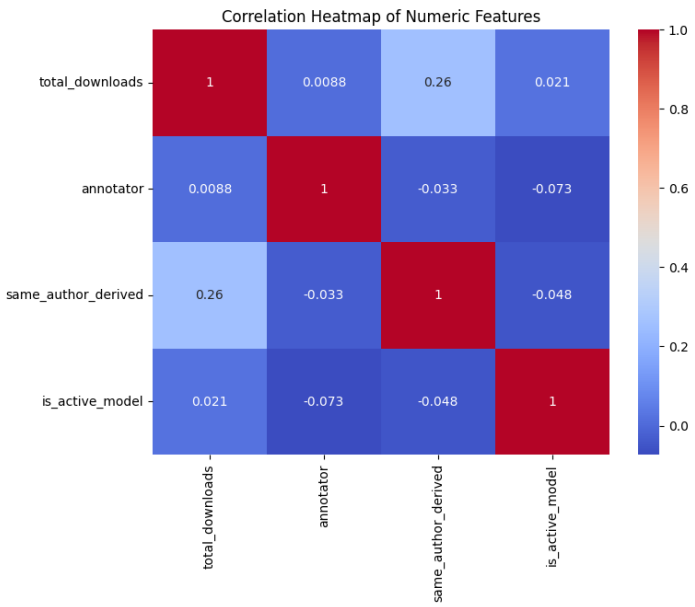
**Downloads vs Type of Users**




**Interpretation:**

- The bar chart shows that models under the Team access tier have the highest average downloads, followed by Enterprise and Free tiers, indicating stronger engagement among collaborative and organizational users.
- In contrast, individual user and Enterprise Plus tiers exhibit lower average downloads, suggesting that structured or paid access environments may drive higher overall model utilization.

**Correlation**



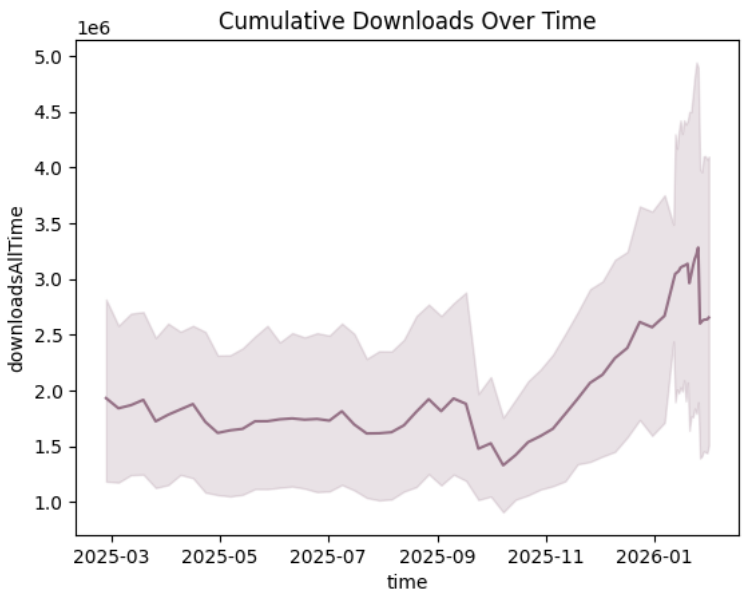

**Interpretation:**

- The correlation heatmap indicates that downloadsAllTime has only a weak positive correlation with same_author_derived (0.26) and almost no correlation with annotator count (0.0088) or active status (0.021), suggesting that total downloads are largely independent of these structural features.
- Overall, the low correlation values across variables imply that model popularity is likely influenced by other external or qualitative factors rather than simple numeric attributes like team size or derivation status.

**Time Series Analysis**

1. **Univariant Analysis**

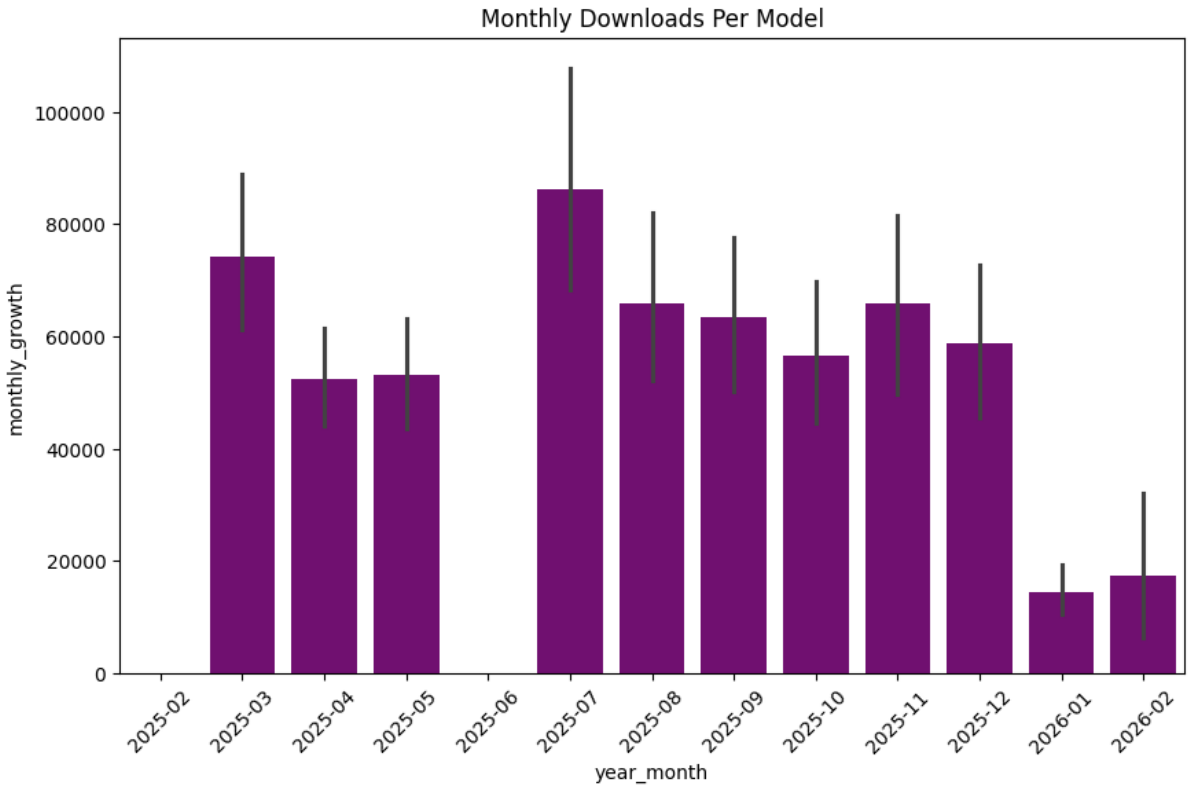
- **Cummulative Download Trend**



**Interpretation:**

The cumulative download trend remains relatively steady from March 2025 to September 2025, indicating a stable growth phase with consistent user adoption. However, a sharp increase around January 2026 suggests a sudden surge in downloads, possibly driven by the release of high-performing models or increased platform traction during that period.

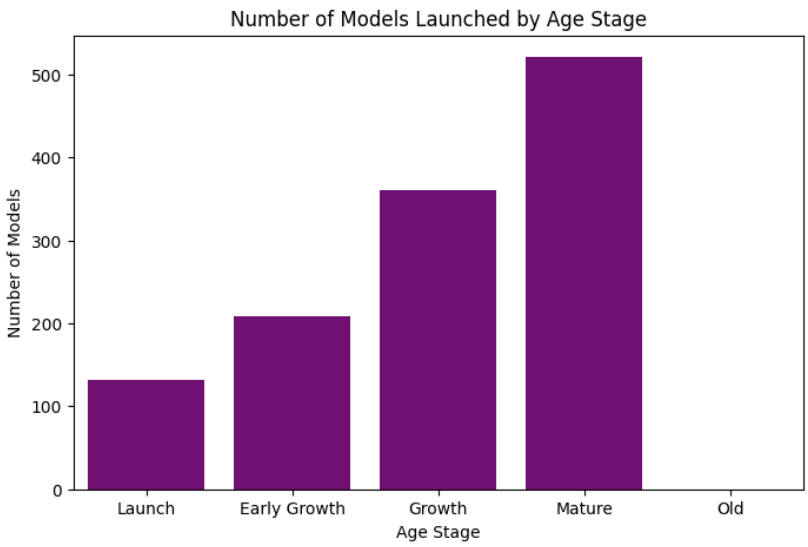
**Monthly Downloads per Model**




**Interpretation:**

- The monthly downloads trend indicates that July 2025 recorded the highest model growth, followed by March 2025 and November 2025, highlighting key periods of elevated user activity.
- These peaks suggest seasonal or release-driven surges in adoption, where specific months experienced significantly stronger download momentum compared to others.

## Number of Models launched based on Age Stage

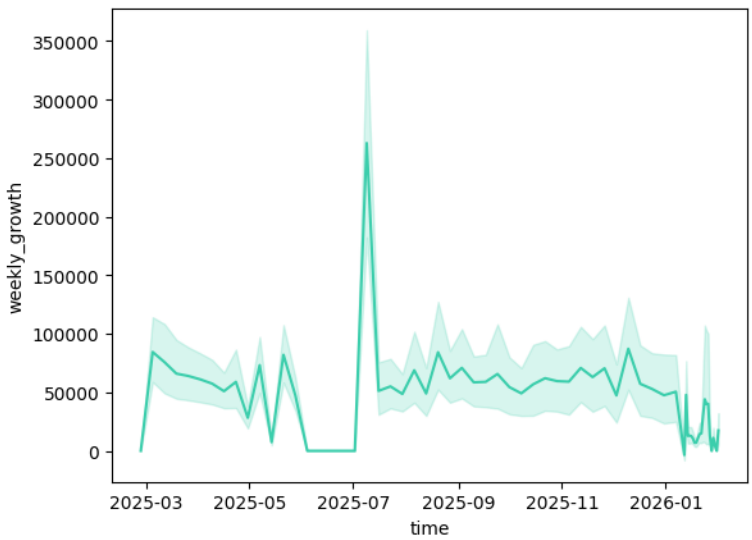



## Interpretation:

- The distribution shows that most models fall into the Mature stage, followed by Growth and Early Growth, indicating that the ecosystem is largely composed of established models rather than newly launched ones.
- The relatively smaller proportion in the Launch stage suggests fewer newly introduced models compared to those that have already progressed through their lifecycle.

## 2.Uni variant Analysis

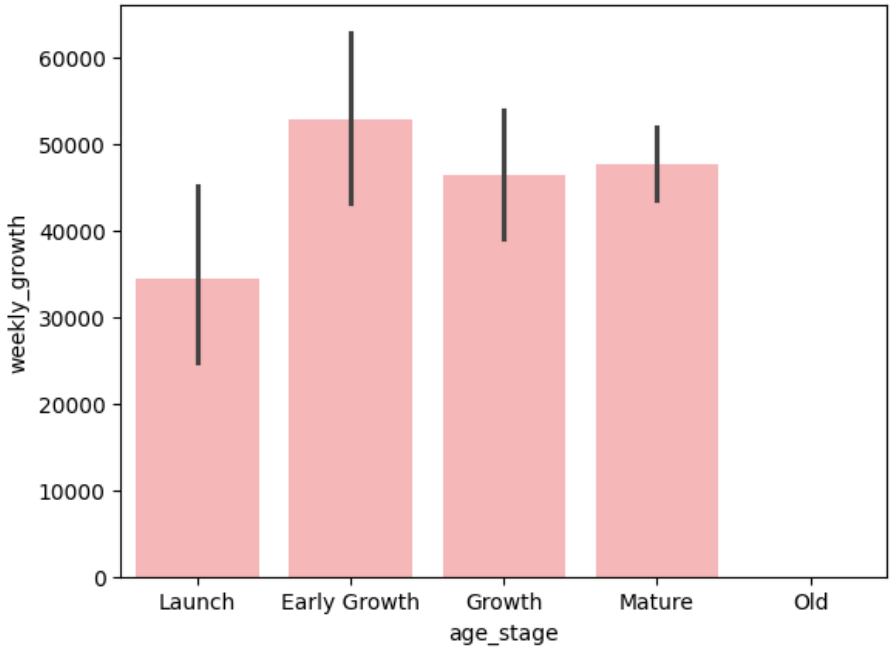
## Time vs Weekly Growth



**Interpretation:**

The weekly growth trend reaches its highest peak around July 2025, indicating a strong surge in short-term adoption during that period. Afterward, growth declines and stabilizes by January 2026, suggesting a transition from rapid expansion to a more steady and consistent growth phase.

**Age Stage vs Weekly Growth**

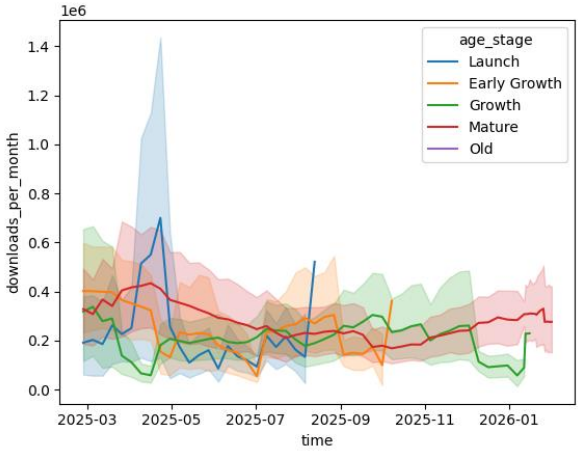



**Interpretation:**

- The bar chart shows that models in the Early Growth stage have the highest average weekly growth, indicating that momentum tends to build shortly after launch rather than immediately at release.
- Mature models still maintain relatively strong growth compared to Launch-stage models, suggesting that sustained adoption can continue well beyond the initial introduction phase.

**3.Multi-variant Analysis**

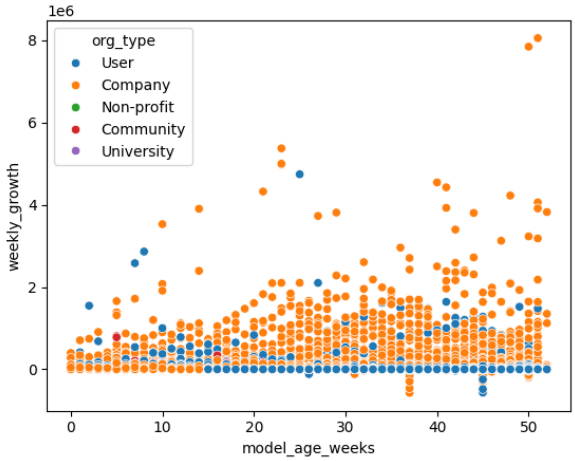
**Time vs Downloads vs Growth**




**Interpretation:**

- The line plot indicates that models in the Launch stage record the highest monthly downloads, suggesting strong initial adoption immediately after release
- However, as models transition into Growth and Mature stages, download momentum becomes more stabilized, reflecting a shift from early hype-driven spikes to sustained usage patterns.

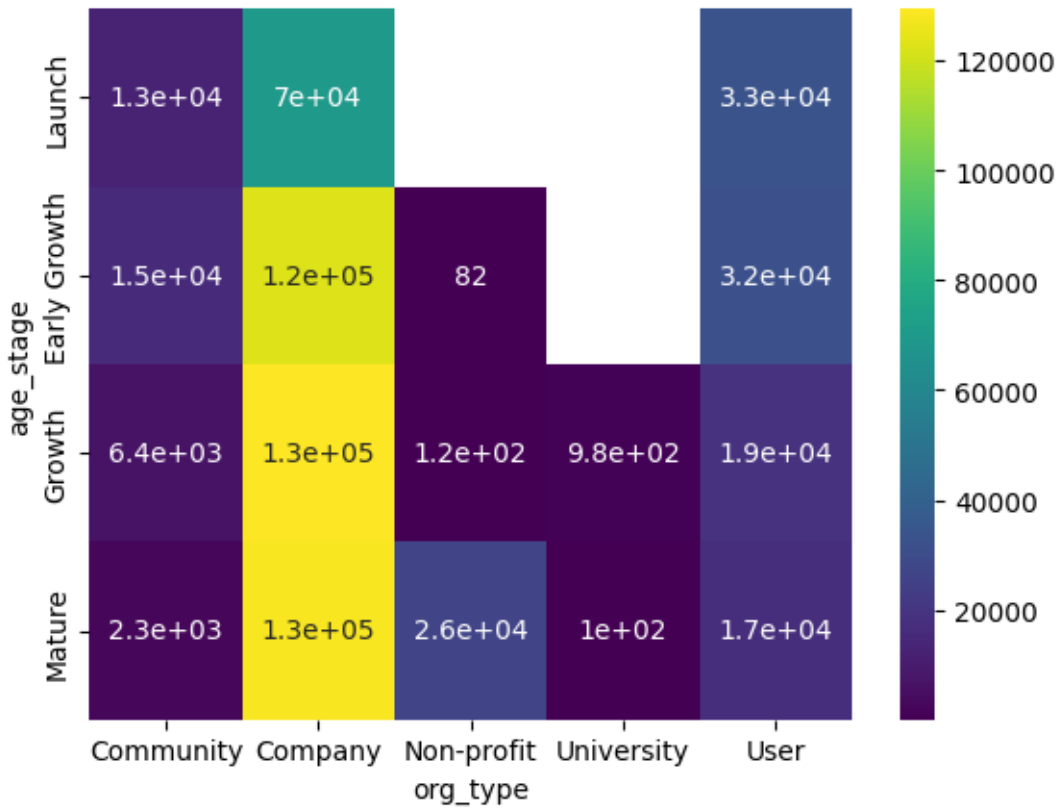
**Age vs Organization Type vs Growth**




**Interpretation:**

- The scatter plot shows that company-developed models exhibit the highest and most concentrated weekly growth, indicating stronger and more sustained performance compared to other organization types.
- In contrast, user-developed models cluster near zero growth, while notable outliers around 50 weeks suggest that some mature company models experience significant growth spikes even at later lifecycle stages.

**Age Stage vs Organization Type vs Average Growth**




**Interpretation:**

- The heatmap shows that company-developed models consistently achieve the highest average weekly growth across almost all age stages, particularly in the Growth and Mature stages, indicating sustained performance beyond initial release.
- In contrast, community, university, and non-profit models exhibit comparatively lower average growth, while user-developed models maintain moderate growth, suggesting that organizational backing—especially from companies—plays a significant role in driving long-term model momentum.

**Summary of Findings:**

**1.Popularity Distribution & Structural Patterns**

- Model downloads follow an extreme right-skewed, heavy-tailed distribution, confirming a winner-takes-all ecosystem where a small number of models dominate total adoption.
- Most models are developed by individual users, yet company-developed models achieve higher median downloads and produce the majority of blockbuster outliers.
- Model inheritance is common (known lineage prevalent), while multimodal input support remains relatively limited, indicating that unimodal architectures still dominate the ecosystem.

**2. Organizational & Geographic Influence**

- Countries such as China, Germany, the United States, and the United Kingdom show the highest average download performance, suggesting strong national AI ecosystems.
- Base models originating from specific countries (notably China) generate higher downstream download impact, highlighting the global influence of foundational models.
- Company-backed models consistently outperform user and non-profit contributions in terms of median downloads, dispersion, and sustained growth.

**3. Feature Relationships & Correlation Insights**

- Downloads show weak correlation with annotator count and derivation status, suggesting that popularity is not strongly driven by simple structural features.
- Moderately sized annotator teams (5–10 contributors) are associated with higher downloads, though extreme outliers indicate that success is multifactorial.
- Multimodal input does not necessarily guarantee higher downloads; unimodal models often achieve stronger adoption across organization types.

**4. Time-Series Growth & Lifecycle Dynamics**

- Download growth exhibits clear surge periods, particularly around mid-2025 and early 2026, indicating release-driven or ecosystem-wide adoption spikes.

- Early Growth stage models show the highest weekly growth momentum, while Mature-stage models maintain steady long-term performance.
- Acceleration distribution peaks at zero, meaning most models experience stable growth, with only a small subset showing extreme volatility or sudden spikes.

**5. Ecosystem-Level Insight**

Overall, the open-source AI ecosystem demonstrates strong popularity concentration, company-driven performance dominance, geographic clustering of success, and lifecycle-dependent growth behavior, where early momentum and strong organizational backing significantly influence sustained model adoption.

**Types of Analysis**

**1.Descriptive Analysis**

Descriptive analysis summarizes the historical and structural characteristics of the AI model ecosystem. In this study, it includes examining the distribution of total downloads, organization type contributions, country-wise model presence, modality usage, and lifecycle stage composition. The analysis revealed an extremely right-skewed download distribution, dominance of individual contributors in model count, and higher median downloads for company-developed models. Time-series summaries further highlighted cumulative download trends, monthly growth peaks, and lifecycle stage distributions (Launch, Growth, Mature).

Examples from this study:

- Distribution of total model downloads
- Organization type frequency counts
- Country-wise average downloads
- Lifecycle stage distribution
- Cumulative and monthly growth trends

**2. Diagnostic Analysis**

Diagnostic analysis investigates the underlying reasons behind observed patterns in model popularity and growth. The analysis identified that company-backed

models show higher median downloads and stronger sustained growth, while unimodal models currently outperform multimodal ones in adoption. Weak correlation between downloads and annotator count suggests that popularity is influenced by qualitative ecosystem factors rather than simple numeric attributes.

Examples from this study:

- Why company-developed models generate higher download outliers
- Why certain countries (e.g., China) show stronger average performance
- Why unimodal models dominate adoption
- Why team size has limited direct impact on popularity

## 3. Predictive Analysis

Predictive analysis focuses on identifying patterns that may forecast future performance trends. Based on lifecycle and growth metrics, models in the Early Growth stage demonstrate the highest weekly growth momentum, suggesting that early-stage acceleration may predict sustained adoption. Time-based trends also show surge periods, indicating potential cyclical or release-driven spikes. Growth stability across mature models suggests that sustained organizational backing may predict long-term success.

Examples from this study:

- Predicting future downloads based on growth acceleration
- Forecasting model performance based on lifecycle stage
- Anticipating high-performing regions or organization types
- Identifying early growth signals of potential breakout models

## 4.Prescriptive Analysis

Prescriptive analysis recommends strategic actions based on analytical insights. The findings suggest that organizational backing, strong base-model lineage, and focused unimodal specialization currently drive higher adoption. Therefore, ecosystem stakeholders may prioritize structured development teams, optimize early-stage growth momentum, and strategically invest in high-performing modalities.

Examples from this study:

- Encourage structured team-based model development
- Strengthen early lifecycle promotion to maximize growth acceleration
- Invest in high-demand modalities (e.g., Text-to-Text, Image-related tasks)
- Leverage successful base-model ecosystems for derivative performance gains

**Future Enhancements**

**1. Incorporate External Ecosystem Signals**
Integrate additional variables such as GitHub stars, model citations, community discussions, or social media engagement to better explain popularity drivers beyond download counts.

**2. Apply Advanced Predictive Modeling**
Develop machine learning models to forecast future downloads, growth acceleration, and lifecycle transitions using historical time-series patterns.

**3. Network & Lineage Analysis**
Perform graph-based analysis on model derivation networks to understand influence propagation, base-model dominance, and knowledge transfer across ecosystems.

**4. Sentiment & Usage-Based Insights**
Include qualitative feedback, user ratings, or deployment case studies to evaluate model impact beyond numerical metrics and better assess real-world adoption quality.

## Conclusion

This study provides a comprehensive analysis of the open-source AI model ecosystem, highlighting strong popularity concentration, company-driven performance dominance, and lifecycle-dependent growth dynamics. The findings reveal that a small number of models capture the majority of downloads, while organizational backing and early growth momentum significantly influence sustained success. Although structural attributes such as annotator count show limited correlation with popularity, geographic origin and base-model lineage

play a meaningful role in downstream impact. Overall, the analysis offers valuable insights into adoption patterns and structural drivers shaping the evolution of AI models within the ecosystem.